

Common PP Errors

[Note: This page is still under construction! It may not be very helpful yet. Please send suggestions of common problems/solutions that aren't on here to Graeme straight away so he can add them!]

This page is an indexed list of common errors that one might see while Post-Processing, and the methods that should work to solve them. If an issue you have isn't on this list, please feel free to add it later once a solution has been identified!

This page is intended to be a **first resort**, a reference guide of common solutions that can be pursued before asking others. There are often multiple solutions to each issue, intended to be tried in order, and this guide is clear as to when it thinks you should escalate your problem to the DS Team Chat or members of other departments.

Cannot find t3st/train files before prepping

```
(full) ppoadmin@ppo-echo:/mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335002_vis/2023-12-10/raw$ ls run*/t3st*glove-or*.h5 | wc -l
ls: cannot access 'run*/t3st*glove-or*.h5': No such file or directory
0
(full) ppoadmin@ppo-echo:/mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335002_vis/2023-12-10/raw$
```

Example: The user is verifying the amount of t3st files with the material type `glove-or`, but the terminal is unable to locate any

When does this happen?: Before prepping files for post-processing, you should always verify that the files are of the correct count, so that you do not accidentally prep an incorrect set of files for later use. This issue occurs when you are performing this verification.

Diagnostic Steps: The first step is to double-check the command you are running. Make sure that you've spelt the material name correctly, your asterisks are appropriately placed, and that you are in the correct directory. If this is all correct, go into some of the `run` folders and inspect the files inside. See if there are named t3st files, or if they are all frames:

```
(full) ppoadmin@ppo-echo:/mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335002_vis/2023-12-10/raw/run_2023-12-10--09-03-16$ ls
frame-1702217004799104-1.h5  frame-1702217039376178-302.h5  frame-1702217074857235-521.h5  frame-1702217110162710-941.h5  frame-1702217150791210-1393.h5  frame-1702217202891843-1747.h5
frame-1702217005925097-3.h5  frame-1702217040506147-317.h5  frame-1702217075761208-629.h5  frame-170221712485311-954.h5  frame-1702217151582189-1300.h5  frame-1702217203334810-1758.h5
frame-1702217006042007-12.h5  frame-1702217041410123-323.h5  frame-1702217076439102-635.h5  frame-170221713163215-960.h5  frame-1702217152838152-1312.h5  frame-1702217204125789-1765.h5
frame-1702217007050041-21.h5  frame-1702217042427007-334.h5  frame-1702217077456163-644.h5  frame-170221714108106-969.h5  frame-1702217153616119-1316.h5  frame-170221720400604-1800.h5
frame-1702217007961017-20.h5  frame-1702217043444070-343.h5  frame-1702217078473137-653.h5  frame-170221714858167-975.h5  frame-1702217157006045-1348.h5  frame-1702217208323650-1811.h5
```

Solution: If there are no named t3st files, renaming has likely not occurred yet. Most often, this is the result of the software team needing more time to rename all of the files - be patient and focus on another day for now. If you know the files are supposed to be renamed, discuss with software to verify this, and escalate further if necessary.

Could not connect to display localhost any x display, qxcbconnection, review_tool.py

```
(full) ppoadmin@ppo-delta:/mnt/data1/u/annotation$ python ~/ppo/tools/visual/review_tool.py post_proc_SFDCUD-IF_acetal-db_2023-12-10_dm21335003_1702664654.txt
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-ppoadmin'
qt.qpa.screen: QXcbConnection: Could not connect to display localhost:12.0
Could not connect to any X display.
(full) ppoadmin@ppo-delta:/mnt/data1/u/annotation$
```

Example: The user is attempting to run review_tool.py, but is prevented from doing so by this error.

When does this happen?: This error is most common when attempting to run review_tool.py, especially within a tmux terminal.

Diagnostic Steps: There are no current diagnostic steps for this issue, as only one solution is known.

Solution: Disconnect from your tmux using `Ctrl+B` and `D`, then kill and re-make your tmux and try again. This issue's cause is not entirely known, but it seems to be related to tmux terminals which are open for an especially long time. If making a new terminal fails to solve the issue, it is time to escalate.

Files Unavailable From a Specific Server

```
(full) ppoadmin@ppo-echo:~$ cd /mnt/ppo-server7
(full) ppoadmin@ppo-echo:/mnt/ppo-server7$ ls
(full) ppoadmin@ppo-echo:/mnt/ppo-server7$
```

Example: The user is attempting to access files stored on `/mnt/ppo-server7/`, but a quick `ls` reveals there are no directories.

When does this Happen?: This happens when first attempting to access a server through MobaX, such as .

Diagnostic Steps: The steps you take depends entirely on where the problem starts. First, ask if you can access the server at all - when you type `ls` in `mnt/ppo-server#`, are there other directories listed? If not, this may mean the server is entirely unmounted.

No module named 'cv2'

```
(base) ppoadmin@ppo-foxtrot:~$ python ~/ppo/tools/visual/review_tool.py
Traceback (most recent call last):
  File "/home/ppoadmin/ppo/tools/visual/review_tool.py", line 3, in <module>
    import cv2
ModuleNotFoundError: No module named 'cv2'
```

Example: The user is attempting to use a tool, but receives this error instead.

When does this happen?: This error can happen when using any of the common tools or their aliases, including: review_tool.py, post_processing.py, vexp/hexp, vroi/hroi, b, or vann/hann.

Diagnostic Steps: There are no diagnostic steps for this issue.

Solution: You silly goose, you forgot to type `aa;caf` before starting your work on your terminal! Don't worry - it happens to all of us - good thing you came to the wiki page first so you can keep this whoopsie as a secret. If for any reason you are already in (full) when you get this error, definitely escalate, as that issue has not happened before.

No Such File or Directory

A terminal window showing a Python script execution. The script is attempting to copy a file from a local path to a remote path. The error message is "FileNotFoundError: [Errno 2] No such file or directory: '/mnt/data/annotation/2023-12-10/2150A6600/t3st+31579_mat+polypropylene_sized-shape+square_colour+white_format+loose_bg+bacon_sku+w4655-w16021_d0+30_height+2_amount-of-fo+20_lane+1_ts+2023-12-10-15-42-07_annotated.h5'". The error is raised in the copyfile function of the shutil module.

Example: The user is attempting to upload, but is stopped by this error.

When does this happen?: This issue happens in a few situations when a user is attempting to access a file, most likely when uploading - but this issue could also happen while using `post_processing.py`.

Diagnostic Steps: Solutions to this problem depends on what function you're attempting to execute, whether that be uploading or prepping a .txt file for post processing.

Solution 1: If you are attempting to upload, this likely means that the .txt file you are uploading with is searching for the appropriate .h5s, but is unable to find them. This could mean that the files have been erroneously moved or deleted - it also could mean the file has already been uploaded. Verify if the error indicates either spec or vis is missing - if just the spec files are missing, there is a chance of recovery. See the **Advanced Solutions** section below. However, if the vis files are missing, it is likely that the files will need to be re-prepped and re-processed.

Solution 2: If you are attempting to prep files, then there likely is some type of typo in your command line. You should always first verify that files are present in the desired location by using `ls run*/[t3st/train]*material*.h5 | wc -l` - assuming this command gives you a count identical to the scans on the spreadsheet, we know the files are present. Double-check your spelling!

Review_tool.py does not open a new window TMUX, not opening, no error

```
(full) ppoadmin@ppp-delta:/mnt/data/annotation$ python ~/ppo/tools/visual/review_tool.py post_proc_SFDCUD_IF_acetal-db_2023-12-10_dm21335003_1902664054.tst
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to /tmp/runtime-ppoadmin
=====TESTING SOME FILE=====
STATUS spec-to-vis_annotatons Auto loading coreg file using serial number 015478 in /mnt/data/vis/annotation/2023-12-10/015478/t3st+31589_mnt+acetal-db_sized-shape+square_colour+blue_format+loose_bg+bacon_s
ku+04656-v16021_dm+30_height+2_amount-of-fo+20_lane+1_ts+2023-12-10--15-51-41_annotated.hs
=====INSTRUCTIONS=====
Review Annotation and ROI pmgs.
Use arrow keys or buttons to navigate between files
Scroll to zoom
Middle click or press z to reset zoom
Annotation: Left click to add or remove annotation boxes
ROIs: Left click to highlight blue and mark ROI to be redone
ROIs: Right click to highlight red and mark ROI to be deleted
Press q to save and exit
Press ! to exit without saving
=====
```

Example: The user is attempting to run review_tool.py, and the command executes successfully, but no additional window opens that can run the actual review tool.

When does this happen?: This issue happens when someone is trying to use review_tool.py, and seems to exclusively happen when in a tmux terminal.

Diagnostic Steps: There are no diagnostic steps for this issue, as there is only one known solution.

Solution: Disconnect from your tmux using **Ctrl+B** and **D**, then kill and re-make your tmux and try again. This is one of a couple errors associated with having tmux terminals open for an extended time - specifically, this error is associated with having a tmux terminal open on two computers (for example, your laptop in the office and your computer at home). If resetting the tmux terminal fails to solve this issue, then it is time to escalate.

Sweeper Path Does Not Appear to Exist calibration, cal_vis, prepping

```
(full) ppoadmin@ppp-foxtrot:/mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335003_spec/2023-12-10/raw$ python ~/ppo/tools/visual/post_processing.py run*/t3st+bone*.hs
ERROR Sweeper Path /mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335003_spec/2023-12-10/cal_vis does not appear to exist.
Traceback (most recent call last):
  File ~/home/ppoadmin/ppo/tools/visual/post_processing.py", line 537, in <module>
    earlier_only=True))
  File ~/home/ppoadmin/ppo/tools/data_utils/timestamp_utils.py", line 277, in get_vis_cal_dir
    verbose=verbose)).parent
  File ~/home/ppoadmin/ppo/tools/data_utils/timestamp_utils.py", line 141, in find_closest_time_match
    key=lambda x: scan_ts - x[1] if scan_ts - x[1] > 0 else float('inf'))
ValueError: min() arg is an empty sequence
```

Example: The user is unable to prep a set of files, as the tool they use will not run. The terminal gives the Error: sweeper Path /mnt/ppo-server2/.../cal_vis does not appear to exist.

When does this happen?: This issue will often happen during the prep of files for post-processing.

Diagnostic Steps: To determine the root of the problem, try and run a prep script on a different local computer and see if you get the same error or if the script runs as expected. Additionally, use the command **mount | grep PPO** to verify that the server the calibrations are stored on is mounted.

Solution 1: If the script fails to run on any local computer, the most likely answer is that the calibrations have not been corrected for yet. This is a step that the software side of PPO is responsible for, just like renaming files. If the files have been renamed but a calibration is not

ready, talk with a member of software (likely Barrett) and ask them to correct the calibrations so that the files can be prepped.

Solution 2: If the script runs normally on other local computers, the prep script may be directed in the wrong location. Take the example of the photo above, and notice that the user has routed into `.../fo/dm21335003_spec/...` - but, while prepping, we should always be routed into the **vis** folder for the respective DM. Reroute to the vis folder and try again. If the script still doesn't work, or if you were already in the vis folder, escalate this issue.

Solution 3: If the server that the calibrations are stored on does not appear to be mounted, you should use the command `mount#` (where the number is the relevant server), along with the password `ppoadmin`. This error can sometimes be the result of the calibration server not being mounted.

Trying to load calibrations from path, but path does not appear to exist

```
(full) ppoadmin@ppo-alfa:/mnt/ppo-server2/RTS/SFDCUD_IF/fo/dm21335003_vis/2023-12-10/raw$ python ~/ppo/tools/visual/post_processing.py run*/t3st*cardboard*.h5
Traceback (most recent call last):
  File "/home/ppoadmin/ppo/tools/visual/post_processing.py", line 561, in <module>
    data_library=ARGOS.data_library
  File "/home/ppoadmin/ppo/tools/visual/post_processing.py", line 130, in __init__
    self.apply_corrections()
  File "/home/ppoadmin/ppo/tools/visual/post_processing.py", line 366, in apply_corrections
    camera_setup=self.spec_dict['camera_setup'][0])
  File "/home/ppoadmin/ppo/tools/chemometric/hyperspec_correct.py", line 325, in correct
    self.try_load_default_cals()
  File "/home/ppoadmin/ppo/tools/chemometric/hyperspec_correct.py", line 160, in try_load_default_cals
    raise ValueError(f'Trying to load calibrations from path {str(camera_path)} but path does not appear to exist.')
ValueError: Trying to load calibrations from path /mnt/ppo-server3/calibrations/Xenics_Cheetah/015478 but path does not appear to exist.
```

Example: The user is attempting to run a prep script, but receives this error, preventing them from continuing.

When does this happen?: This occurs when attempting to run a prep script for files during post-processing.

Diagnostic steps: Use the command `mount | grep PPO` and see if the server that the calibrations are pathed to is mounted. To figure out where the calibrations are, look at the last line of the error: note in the example above, the calibrations are found in **ppo-server3**. Check to see if the server is mounted, either as `IO` or as `IW`.

Solution 1: If the server the calibrations are held on is not mounted, use the command `mount#` (using the server's number instead of #). The terminal will ask you for a password: enter `ppoadmin` and this will confirm the mounting of the server into Read-only. This should enable your local computer to find the path to the calibrations.

Solution 2: If the server is already mounted, or solution 1 doesn't work, it may be an issue with your tmux window. Close and kill your tmux and re-open, trying again in the new terminal. If this doesn't work either, it is time to escalate the issue.

[Placeholder] Incorrect permissions to upload (error unknown)

When does this happen?: This issue occurs as you're trying to upload, and will prevent you from beginning the upload whatsoever.

Diagnostic Steps: To verify, use the command `mount | grep PP0` to verify that the server you are uploading too has been given read/write permissions (next to the server ID, it will read 'rw'.)

```
(full) ppoadmin@ppo-delta:/mnt/data1/u/annotation$ mount | grep PP0
//ppo-server2.local/data1 on /mnt/ppo-server2 type cifs (rw,relatime,vers=1.0,cache=strict,soft,nounix,serverino,mapposix,noperm,rsize=61440,wsiz=65536,echo_interval=60,actimeo=1)
(full) ppoadmin@ppo-delta:/mnt/data1/u/annotation$
```

You can see here: This server, ppo-server2, is mounted in the 'rw' format, as is visible in the command line.

Solution: The server is likely mounted incorrectly in the `read only` configuration, rather than `read/write`. To correct this, you can use the command `mount#w [username]`, where # represents the server number you wish to mount. This command *only* works if you have permission to do this; your username would be allocated to you when you gain permission to change the server to read/write. If you don't have permission to do this yet, send a message to Alan Roofs on the software team.

If the server is already mounted in read/write, this may be a different and unknown issue - it is time to escalate.

Advanced Solutions

These are 'next resort' solutions which may function for specific problems, but can't fit neatly within a solution box up above! Best referred to if a problem tells you to consider an advanced solution.

Vis .h5 files available, but spec files are missing

If, while attempting to upload t3st files, the vis files are available - but the spec files are missing - this is possibly due to the deletion of some of the files erroneously - possibly by another uploader, or possibly due to some other issue. This would best be demonstrated by the absence of files within the spec folder, but the clear presence of files within the vis folder.

Due to the way that our text files work, it is possible to transform the coordinates of the vis masks onto a set of spec files, and so spec files are not technically necessary. If done properly, you can avoid needing to completely redo the set of files. Follow the steps below in order:

1. **Move the vis files** from their current folder, usually

`/mnt/data1/u/annotation/[vis camera]` , and place them into a temporary placeholder folder, such as `/mnt/data1/u/annotation/temp` . The prep tool will skip writing any new files if it detects the vis files as present, so we need to trick the software into thinking it is writing a brand-new file. You can use the command `mv *files* /mnt/data1/u/annotation/[temp folder name]` to do so.

2. **Move the .txt file** into this temporary folder as well. These files are timestamped, and so it is possible to distinguish the new .txt file that the prep script will create from this one - but, it is easier to identify later if it is separated to begin with.
3. **Re-run the prep script** on the appropriate files in the server. Since we've moved the relevant files, the prep script will re-write a set of both vis and spec .h5s into the relevant folders, and will re-create a new text file as well.
4. **Remove the new vis and .txt files** from the local computer using the `rm` command. The new text file fails to contain the information that the old one does, and using it will effectively require re-doing the entire set of files. Once you've removed the text file, go into the vis folder and remove the newly-minted vis files, as these also encode incorrect information.
5. **Move the files in the temp folder** back into their appropriate places: the .txt file in `/mnt/data1/u/annotation` , and the vis files into `/mnt/data1/u/annotation/[vis camera]` .
6. **Run the text file through the review tool:** use the tag `--make_extracts` but avoid using `--spec` for now. Currently, the vis files will have retained the way they were originally post-processed, but the spec files are new and need to be written over. To guarantee this happens, we need to edit every single vis mask file, so that when extracts are written, new files will write over our incorrect spec files. To do this, click to remove and then click to replace a single annotation box for each scan. Don't move the annotation box - essentially, double-click in the same location. This will signal to the command that the annotations have been altered, even though their position is the same, which will trigger it to re-write.
7. **Re-review the spec extracts:** The spec extracts will be transformed from the vis locations. Because of this, where the transformation failed before (and accidentally showed part of an o-ring or cut off the FO), it will do so again. Re-review the spec extracts and change the

ones which are cut off by using `review_tool.py` with the tags `--spec --make_extracts` .

8. **Upload the new files** back up to the server now that they are verified to be done properly.
9. **Clean up** by deleting the uploaded files after upload is complete. Additionally, make sure you've removed the temporary folder that you set up.